PCBC H Parametric & Sequences Test (Practice A) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A football is kicked from a height of 2 feet at an angle of 38° with an initial speed of 64 ft/s.

a) Write a set of parametric equations for the motion of the football. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the football.

c) Determine how long the football is in the air.

d) Determine how far the football traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, -\frac{4}{81}, \cdots$

Write out the sum. (3 points)

9. $\sum_{k=1}^{n} \left(k^2+1\right)$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{40} (k+3)$

11. $\sum_{k=1}^{30} \left(2k^2-5k\right)$

12. $\sum_{k=9}^{24} 3k^3$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x+2)^6$

14. $(2x-5)^4$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Express the sum using summation notation.

E1. $5+8+11+14+\cdots+62$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $\frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \cdots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Football: $h = 2$, $\theta = 38°$, $v = 64$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (64\cos 38°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (64\sin 38°)t - 16t^2 + 2$ — *+h: kicked from 2 ft, so +2*

$x_1(t) = 64\cos(38°)\cdot t,\ \ y_1(t) = 64\sin(38°)\cdot t - 16t^2 + 2$ — *set up: what you type in*

$0 \le t \le 3$, tstep 0.01; x : 0 to 140, y : 0 to 30 — *if the graph doesn't come back down, raise tmax*

b) $\boxed{y \approx 26.258 \text{ ft}}$ — *trace to the top of the path ($t \approx 1.231$); not Analyze, Maximum*

c) $\boxed{t \approx 2.512 \text{ s}}$ — *trace to where $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 126.707 \text{ ft}}$ — *same point, read x. trace landing on $t = 2.512$ shows ≈ 126.687; a trace answer like that is fine*

*Watch: answers in **feet**, the speed was ft/s.*

8. $\frac{1}{\frac{1}{3}}, -\frac{2}{\frac{2}{9}}, \frac{3}{\frac{3}{27}}, -\frac{4}{\frac{4}{81}}, \ldots$

top: n, bottom: 3^n — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive, so n + 1*

$\boxed{a_n = (-1)^{n+1}\frac{n}{3^n}}$ — *check $n = 1$: $\frac{1}{3}$*

9. $\sum_{k=1}^{n}(k^2+1)$

$(1^2+1) + (2^2+1) + (3^2+1) + \cdots + (n^2+1)$ — *$k = 1,2,3$, then the last term with n*

$\boxed{2 + 5 + 10 + \cdots + (n^2+1)}$ — *don't add*

10.

$\sum_{k=1}^{40} k + \sum_{k=1}^{40} 3$ — *(2) split the sum*

$\frac{40(41)}{2} + 3(40)$ — *(6) and (5) with $n = 40$; calculator after this line*

$\boxed{940}$

11.

$2\sum_{k=1}^{30} k^2 - 5\sum_{k=1}^{30} k$ — *(3) split, (1) constants out*

$2\left[\frac{30(31)(61)}{6}\right] - 5\left[\frac{30(31)}{2}\right]$ — *(7), (6) with $n = 30$*

$\boxed{16585}$

12.

$3\sum_{k=9}^{24} k^3$ — *(1) constant out*

$3\left[\sum_{k=1}^{24} k^3 - \sum_{k=1}^{8} k^3\right]$ — *(4) lower index is 9, not 1: $j=8$*

$3\left[\left(\frac{24(25)}{2}\right)^2 - \left(\frac{8(9)}{2}\right)^2\right]$ — *(8): square the whole bracket*

$\boxed{266112}$

13. $(x+2)^6$: $n=6$, row 1 6 15 20 15 6 1. The sign goes with its term.

$1x^6(2)^0 + 6x^5(2)^1 + 15x^4(2)^2 + 20x^3(2)^3$ — *blanks first: first term's powers count down, second term's count up*

$+15x^2(2)^4 + 6x^1(2)^5 + 1x^0(2)^6$

$\boxed{x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64}$ — *second term positive: every term positive*

14. $(2x-5)^4$: $n=4$, row 1 4 6 4 1. The sign goes with its term.

$1(2x)^4(-5)^0 + 4(2x)^3(-5)^1 + 6(2x)^2(-5)^2 + 4(2x)^1(-5)^3$ — *blanks first: first term's powers count down, second term's count up*

$+1(2x)^0(-5)^4$

$(2x)^4 = 16x^4, \ldots$ — *the whole first term gets the power*

$\boxed{16x^4 - 160x^3 + 600x^2 - 1000x + 625}$ — *second term negative: odd powers of -5 are negative, so signs alternate*

Extra practice (not scored)

E1. $5+8+11+\cdots+62$

up by 3: $3k+2$ — *$k=1$ gives 5*

$3k+2=62 \Rightarrow k=20$ — *solve for the upper index*

$\boxed{\sum_{k=1}^{20}(3k+2)}$

E2. $\frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{5}{11}, \ldots$

top: $2n+1$, bottom: $(n+1)^2$ — *write 1 2 3 4 above; top and bottom separately*

$\boxed{a_n = \frac{2n+1}{(n+1)^2}}$ — *odd tops from 3; squares from 4. Check $n=1$: $\frac{3}{4}$*